

Causal Analysis of Attention Heads in GPT-2 Small

Activation Patching and Ablation for IOI

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1 Overview

This project investigates how GPT-2 Small internally solves a simple **Indirect Object Identification (IOI)** task. The goal was not to reproduce the complete IOI circuit from prior work, but to build a compact causal analysis pipeline and identify internal components that measurably affect the model’s behavior.

The analysis used three stages:

1. residual-stream activation patching to localize the relevant layers;
2. attention-head activation patching to identify candidate heads;
3. attention-head ablation to test whether removing those heads changes the model’s behavior.

All experiments used GPT-2 Small with no training or fine-tuning.

2 Dataset and Behavioral Metric

I generated 1,000 synthetic IOI examples using several controlled sentence templates and a fixed set of names. Each example contains a clean and a corrupted version.

Example:

Clean:

When Carol and Frank arrived at school,
Frank showed the picture to

Corrupted:

When Frank and Carol arrived at school,
Carol showed the picture to

For the clean prompt:

clean correct = Carol, clean incorrect = Frank.

For the corrupted prompt, the roles are reversed:

corrupted correct = Frank, corrupted incorrect = Carol.

For all experiments I used a fixed clean-vs-corrupted logit axis:

$$LD = \text{logit}(\text{clean correct}) - \text{logit}(\text{clean incorrect}).$$

Therefore:

$$LD > 0$$

indicates preference for the clean answer, while

$$LD < 0$$

indicates preference for the corrupted answer.

This fixed direction is important because it allows clean, corrupted, and intervened runs to be compared using the same behavioral metric.

3 1. Residual-Stream Activation Patching

For each layer, the clean residual activation was inserted into the corrupted forward pass at the final prediction position.

The normalized patching score was

$$P = \frac{LD_{\text{patched}} - LD_{\text{corrupt}}}{LD_{\text{clean}} - LD_{\text{corrupt}}}.$$

A value near 1 means that the intervention restores most of the clean-corrupted behavioral gap.

The main result is shown below.

Table 1: Mean residual-stream patch effect over 1,000 examples.

Layer	Mean Patch Effect
0	0.000
1	-0.000
2	0.000
3	0.000
4	-0.000
5	-0.001
6	-0.000
7	0.003
8	0.109
9	0.265
10	0.528
11	0.838

The transition is clear: patching early layers has almost no average effect, while Layers 8–11 show a sharp increase. Layer 11 has a mean normalized patch effect of 0.838.

This motivated restricting the next analysis to Layers 10 and 11.

4 2. Attention-Head Activation Patching

GPT-2 Small has 12 attention heads per layer, so Layers 10 and 11 contain 24 candidate heads.

Instead of patching the complete residual vector, I patched the output of a single attention head using:

```
blocks.L.attn.hook_z
```

The strongest candidates were:

L10H0 is the strongest candidate by average patch effect. L10H6 and L10H1 are the next strongest.

An interesting case is L11H11: its average effect is smaller, but it has a positive patch effect for every example. This shows why both magnitude and consistency are useful when selecting candidate heads.

5 3. Attention-Head Ablation

To test whether candidate heads contribute to the corrupted computation, each head was set to zero at all token positions:

```
activation[:, :, head, :] = 0
```

Table 2: Leading heads by mean patch effect.

Head	Mean	Median	Positive Fraction
L10H0	0.101	0.088	0.985
L10H6	0.068	0.043	0.995
L10H1	0.056	0.037	0.920
L10H10	0.050	0.051	0.909
L11H1	0.036	0.022	0.862
L11H3	0.032	0.022	0.956
L11H11	0.032	0.026	1.000
L10H9	0.013	0.009	0.943
L11H9	0.013	0.007	0.799
L11H0	0.009	0.007	0.941

The ablation effect was defined as

$$A = LD_{\text{ablated}} - LD_{\text{corrupt}}.$$

Because the corrupted model is normally negative on the clean-vs-corrupted axis:

- $A > 0$: ablation moves the prediction toward the clean answer;
- $A < 0$: ablation moves the prediction further toward the corrupted answer.

The largest average effects were:

Table 3: Head ablation results.

Head	Mean	Median	Positive Fraction
L11H0	0.544	0.536	0.876
L10H0	0.338	0.285	0.778
L10H6	0.318	0.219	0.816
L11H1	0.091	0.061	0.615
L10H10	0.087	0.071	0.603
L11H3	0.031	0.029	0.556
L11H11	0.023	0.023	0.562
L10H1	0.020	-0.042	0.428

The ablation ranking is not identical to the patching ranking. For example, L10H0 is the strongest head by patching, whereas L11H0 has the largest mean ablation effect.

This is expected because the two interventions answer different questions. Patching asks whether a clean internal activation can restore behavior in a corrupted computation. Ablation asks how the computation changes when a component is removed.

6 Analysis

The most robust finding is the **late-layer localization**. Residual patching shows almost no average effect before Layer 8, followed by a strong increase through Layers 8–11. This suggests that the model’s final IOI decision is strongly dependent on late residual representations.

The head-level analysis provides a more specific localization. A small subset of heads in Layers 10 and 11 consistently produces larger patch effects than the other tested heads. In particular, L10H0 and L10H6 show both relatively large mean effects and high consistency across the dataset.

The ablation results provide complementary evidence but should not be treated as a direct ranking of “importance.” A positive ablation effect means that removing the head moves the corrupted model toward the clean answer; a negative effect means that removal moves it further toward the corrupted answer. Therefore, the sign describes the direction of behavioral change.

The results support the conclusion that IOI behavior is concentrated in a small number of late-layer components, but they do **not yet identify the complete IOI circuit**. A high patching effect alone does not tell us what a head is computing or which tokens it reads.

7 Limitations and Next Step

The dataset is synthetic and uses a restricted set of sentence templates. The corruption mechanism is also simpler than the full IOI benchmark used in the literature.

The head analysis was restricted to Layers 10 and 11 based on the residual patching results. This is computationally efficient but could miss weaker components in earlier layers.

Most importantly, the current analysis does not yet determine *what the candidate heads do*. The next useful experiments are therefore:

1. head $\times$ token-position patching;
2. visualization of attention patterns for the strongest heads;
3. clean-prompt ablation;
4. controlled combinations of the strongest candidate heads.

8 Conclusion

Using only inference-time causal interventions, the project localized IOI behavior in GPT-2 Small from the residual stream down to individual attention heads.

The strongest result is the layer-level transition:

Layer 7: 0.003 $\rightarrow$ Layer 8: 0.109 $\rightarrow$ Layer 9: 0.265 $\rightarrow$ Layer 10: 0.528 $\rightarrow$ Layer 11: 0.838.

At the head level, L10H0 and L10H6 are the strongest patching candidates, while L11H0 produces the largest average ablation effect.

Overall, the experiments provide evidence for a compact late-layer mechanism rather than uniformly distributed computation. The next stage is to determine the information flow and functional roles of the candidate heads.